



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Innovative Digital Documentation Framework for Heritage Education and Outreach : Case study from India

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Abstract

India has a vast amount of built heritage. According to a report by a think tank of Government of India - Niti Aayog, it has over 5,00,000 heritage structures and sites. Yet only about 100,000 are officially listed, and a mere 7% enjoy legal protection.

With the advancements in various digital technologies, there is an immense scope of documentation of these built heritage fabric. This paper attempts to bridge this gap by leveraging the financial and physical accessibility of new technologies and the integration of LLM and ML, in the field of cultural heritage. It attempts to harness the potential of cross collaborations amongst various disciplines to enhance the findability, accessibility and interoperability of the once stagnant data of the cultural heritage.

This paper highlights a case study of UNESCO World Heritage Volunteers 2025 camp from Vadnagar titled 'Vadnagar Chronicles: Unveiling The Layers Of A Historic Town'. This 7 day workshop, consisting of 10 young participants, focused on enhancing the volunteers' skills as well as addressing the accessibility and innovation in the documentation data. The workshop brought together participants from diverse disciplines, including photography, architecture, computer engineering, design, civil engineering, museology, and art conservation. One segment of the workshop leveraged photogrammetry, a digital documentation tool to document the Kirti Toran (an ornamental gateway), an important architectural feature of the city, surviving through centuries.

The methodology adopted a bi fold approach focusing on theoretical as well as practical methods. The foundations of photogrammetry such as camera geometry, image overlap requirements, lens calibration, ground sampling distance (GSD), sources of reconstruction error etc was introduced in the theory sessions; followed by onsite preparation of a documentation strategy identifying the sequence of image capture, optimal camera

positions, areas susceptible to shadows or occlusions etc. In order to understand the accessibility of different softwares, the acquired datasets were processed using Agisoft Metashape Professional and RealityScan (Epic Games). The 3D modelling included image alignment, dense point cloud reconstruction, mesh generation etc.

Following post-processing and quality review, the successfully generated 3D model and associated documentation were uploaded on Wikimedia Commons enhancing the findability, accessibility, interoperability, and reusability (FAIR) of the documentation by making it freely accessible.¹ However, due to time and budget constraints, drone imagery could not be acquired, resulting in incomplete coverage of inaccessible upper sections. Additionally, as Wikimedia Commons currently supports only STL files for 3D models, the textured rendering could not be retained.²

This interdisciplinary collaboration generated new ideas for transforming the 3D documentation into immersive educational and heritage tourism experiences for wider audiences. While the workshop discussed new concepts, there was insufficient time to fully develop this framework. This work will continue in the upcoming camps as part of a long-term research initiative focused on building a digital heritage archive and exploring new applications.

References

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